

A Geometric View of the Hausman Test

Research note for POE5 section 15.4.2

1. The puzzle

To test whether the random effects (RE) estimator is consistent, we compare it with the fixed effects (FE) estimator. For a single coefficient β_k the Hausman statistic is

$$t = \frac{b_{FE,k} - b_{RE,k}}{[\widehat{\text{var}}(b_{FE,k}) - \widehat{\text{var}}(b_{RE,k})]^{1/2}}.$$

The denominator is surprising. In general,

$$\text{var}(b_{FE,k} - b_{RE,k}) = \text{var}(b_{FE,k}) + \text{var}(b_{RE,k}) - 2\text{cov}(b_{FE,k}, b_{RE,k}),$$

but Hausman showed that under the null hypothesis

$$\text{cov}(b_{FE,k}, b_{RE,k}) = \text{var}(b_{RE,k}),$$

so that

$$\text{var}(b_{FE,k} - b_{RE,k}) = \text{var}(b_{FE,k}) - \text{var}(b_{RE,k}). \quad (15.37)$$

Why should the covariance between two different estimators equal the variance of one of them? Geometry gives a clean answer.

2. Estimators as vectors

Estimators are random variables, and random variables with finite variance form a vector space with the inner product

$$\langle X, Y \rangle = \text{cov}(X, Y), \quad \|X\|^2 = \text{var}(X).$$

In this space, *length* is standard deviation and *orthogonality* is zero covariance.

Under H_0 (the individual effects u_i are uncorrelated with the regressors), both b_{FE} and b_{RE} are consistent for β . Treating them as (asymptotically) unbiased, the squared distance from the true value is simply the variance:

$$\|b - \beta\|^2 = E[(b - \beta)^2] = \text{var}(b).$$

So “closer to β ” means “smaller variance”. Under H_0 , b_{RE} is the (asymptotically) efficient GLS estimator, so it is the closest point to β among the consistent estimators we consider. b_{FE} is consistent but further away.

3. The line of consistent estimators

Let $d = b_{FE} - b_{RE}$ and consider the family

$$b(\lambda) = b_{RE} + \lambda d = (1 - \lambda) b_{RE} + \lambda b_{FE}, \quad \lambda \in \mathbb{R}.$$

Every member is consistent under H_0 . This family is the straight line through b_{RE} and b_{FE} in Panel A. Its squared distance from β is

$$\text{var}(b(\lambda)) = \text{var}(b_{RE}) + 2\lambda \text{cov}(b_{RE}, d) + \lambda^2 \text{var}(d).$$

Because b_{RE} is efficient, no point on the line can be closer to β , so this quadratic must be minimised at $\lambda = 0$. Setting the derivative to zero there gives

$$\text{cov}(b_{RE}, b_{FE} - b_{RE}) = 0.$$

Geometric reading. The circle in the figure contains all estimators with the same variance as b_{RE} . If the line entered the circle, some combination of FE and RE would beat RE, contradicting efficiency. The line can therefore only *touch* the circle at b_{RE} : it is a tangent, and a tangent is perpendicular to the radius. The radius is $b_{RE} - \beta$, and the tangent direction is $b_{FE} - b_{RE}$; the right angle between them is marked in both figures.

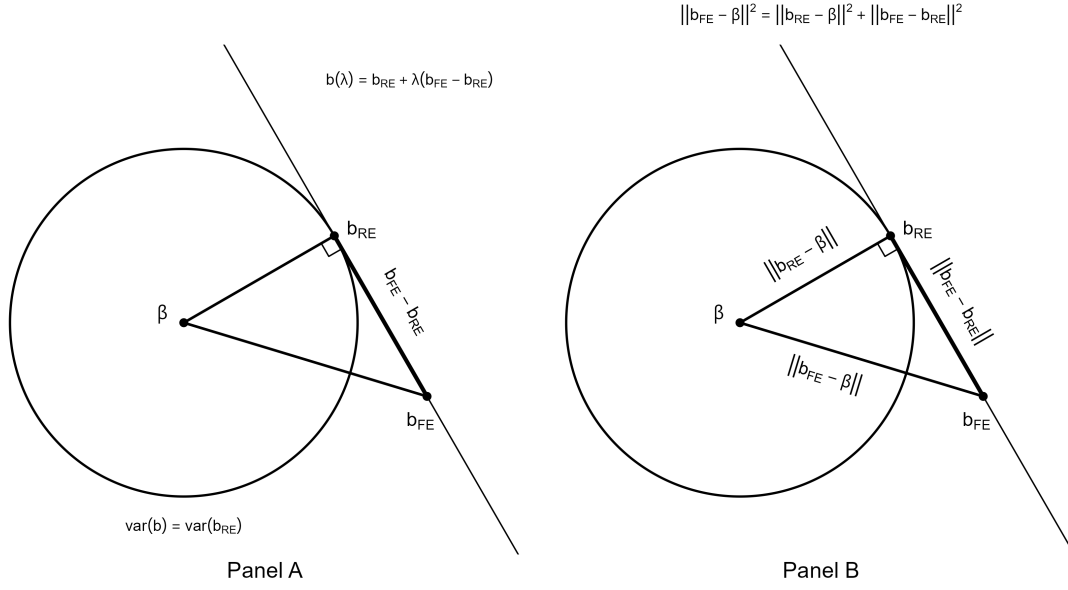


Figure 1: Panel A: the line of consistent estimators is tangent to the variance circle at b_{RE} . Panel B: the resulting right triangle.

4. Pythagoras gives Hausman's result

The orthogonality $\text{cov}(b_{RE}, d) = 0$ immediately gives the covariance result:

$$\text{cov}(b_{FE}, b_{RE}) = \text{cov}(b_{RE} + d, b_{RE}) = \text{var}(b_{RE}) + \underbrace{\text{cov}(d, b_{RE})}_{=0} = \text{var}(b_{RE}).$$

It also gives a right-angled triangle with vertices β , b_{RE} and b_{FE} (Panel B). By the Pythagorean theorem,

$$\|b_{FE} - \beta\|^2 = \|b_{RE} - \beta\|^2 + \|b_{FE} - b_{RE}\|^2,$$

which in variance language reads

$$\text{var}(b_{FE}) = \text{var}(b_{RE}) + \text{var}(b_{FE} - b_{RE}).$$

Rearranging gives (15.37). The hypotenuse (FE) is the long side; the efficient estimator (RE) and the contrast (FE - RE) are the two legs.

5. What this buys us, and what it requires

No covariance to estimate. We never need $\text{cov}(b_{FE}, b_{RE})$; the two separately reported variances are enough.

A non-negative denominator, in principle. The hypotenuse is always at least as long as a leg, so $\text{var}(b_{FE}) \geq \text{var}(b_{RE})$ in population. In finite samples the *estimated* difference can still be zero or negative, in which case the simple statistic cannot be computed.

Power under the alternative. If u_i is correlated with the regressors, b_{RE} is inconsistent: it no longer sits near β , while b_{FE} still does. The contrast $b_{FE} - b_{RE}$ then does not vanish as N grows, and the test rejects. This is the test working as intended, not a failure of the geometry; the variance formula is only needed to get the null distribution right.

The key assumption is efficiency. The tangency argument needs b_{RE} to be *fully* efficient under H_0 . If the idiosyncratic errors are heteroskedastic or serially correlated beyond the RE structure, RE is not the closest point, the line can cut through the circle, orthogonality fails, and (15.37) no longer holds. In that case, use a cluster-robust version of the test, for example the Mundlak auxiliary regression or a bootstrap of the contrast.

Several coefficients. The argument holds for any linear combination $c'b$, so it extends to the vector case:

$$H = (b_{FE} - b_{RE})' [\widehat{V}_{FE} - \widehat{V}_{RE}]^{-1} (b_{FE} - b_{RE}) \stackrel{a}{\sim} \chi_K^2 \quad \text{under } H_0.$$